The Complete Treatise on Actually

A Unified Theory of
Verifiable Value, Informational Symmetry,
Legitimate Decision, and
Residual Risk Allocation

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Abstract

This treatise develops *Actually* as a jointly produced institutional technology for converting economic observations into verifiable representations of value, investigating and resolving material informational asymmetries, validating decisions under ethical and legal constraints, and allocating uncertainty that remains after investigation. The construction unifies three complementary foundations: the stochastic ninefold decomposition and conservation laws of Value-Based Accounting; the programmatic investigation, decision-enablement, and legitimacy conditions of Asymmetric Informational Ethics (AIE); and the individual, actuarial, systemic, aggregate, and model-risk structure of the General Theory of Insurance. The motivating problem is the Egyptian–Roman distinction in *Hinduism continues*: one economy has not produced accounting, another has, yet accounting itself can create an informational asymmetry that prevents their joint long-run development. The present work proposes a technical resolution without treating accounting, disclosure, or insurance as individually sufficient.

The formal system contains a value-state vector, a material-asymmetry field, a set of outstanding decisions, a residual-risk functional, and a governance record. We derive transaction-level conservation, a strengthened finite-program convergence result, a Bayesian value-of-information result, and a risk-allocation identity based on positive homogeneity. We distinguish local program termination from global social termination, and informational symmetry from omniscience. The treatise then develops financial-planning applications, state-space estimation, econometric identification, legal governance, interdisciplinary correspondences, cybersecurity controls, and a reproducible algorithmic architecture. Mathematical results are stated with their assumptions; analogies across physics, chemistry, biology, linguistics, psychology, religion, and political science are explicitly distinguished from mathematical equivalences. Actually is not asserted to eliminate uncertainty. Its purpose is to make material uncertainty visible, investigated, legitimately actionable, and fairly allocated.

The treatise ends with "The End"

Keywords: actually; value-based accounting; asymmetric informational ethics; insurance; information economics; accounting conservation; legitimacy; systemic risk; epistemic symmetry; financial planning; econometrics; auditability; cybersecurity; institutional design.

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1 Introduction

1.1 The motivating problem

The starting point is a distinction made in *Hinduism continues*. In that text, an "Egyptian economy" is a psychological economy that has never produced accounting, while a "Roman economy" is one that has. The text then makes the more important observation: the two economies possess asymmetric information, and the production of accounting is itself implicated in the asymmetry between them. Neither reaches the long run merely by possessing or lacking accounting. They must jointly produce a further technology that removes the asymmetry generated through accounting; that technology is called *actually* [1].

The labels "Egyptian" and "Roman" are technical labels internal to that conceptual manuscript. They are not historical, ethnic, national, or civilizational classifications in this treatise. The substantive problem is general: one participant possesses a formal accounting representation that another participant cannot inspect, interpret, verify, or contest on equal terms. A ledger may be internally balanced and still fail as a common instrument of understanding. A disclosure may be extensive and still omit material context. A contract may be formally valid and still distribute uncertainty in a way that affected parties could not reasonably assess.

The problem is therefore not solved by accounting alone, disclosure alone, or insurance alone. The three are complementary. The information-economics literature has long shown how asymmetric information can generate adverse selection, signalling equilibria, and credit rationing; those mechanisms make institutional design central rather than incidental [5, 43, 45]. Accounting supplies a structured representation of value. Investigation supplies a process for resolving relevant differences in knowledge. Insurance supplies a way to allocate losses and uncertainty that cannot be eliminated by information. The central thesis is that Actually is the institutional technology that connects these functions.

1.2 Three foundations and one synthesis

The first foundation, *The Theory of Value-Based Accounting*, replaces a purely static reading of the identity between assets, liabilities, and equity with a stochastic decomposition of value trajectories. It partitions each register into increasing, constant, and decreasing components and derives a conservation law, a drift-balance relation, and a scalar measure of value momentum [2].

The second foundation, *The Complete Treatise on Asymmetric Informational Ethics*, treats ethical activity as a collection of programs initiated by information asymmetries, developed through iterative investigation, and closed only when relevant asymmetries and outstanding decisions have both been dealt

with. It adds a three-component legitimacy condition based on inviolable constraints, revealed values, and ideal endorsement under full information [3].

The third foundation, *A General Theory of Insurance*, treats insurance as a mechanism for reducing and allocating uncertainty across individual, actuarial, systemic, aggregate, and model dimensions. It motivates premiums and capital allocations by the marginal contribution of exposures to a global risk functional [4].

Foundation	**Primary object**	**Question answered**
Value-Based Accounting	Value-state vector and transaction flows	What is happening to assets, liabilities, and equity, and do the reported flows conserve the accounting identity?
Asymmetric Informational Ethics	Agents, asymmetries, investigation programs, and decisions	What do affected agents know, what remains materially different, and which decisions are legitimate?
General Theory of Insurance	Losses, uncertainty functionals, risk pools, and capital	Which risks remain after investigation, who bears them, and how should they be financed?
ACTUALLY	Integrated institutional protocol	How can value be represented, mutually understood, legitimately acted upon, and protected against residual uncertainty?

Table 1: The complementary foundations of Actually.

1.3 Contributions and epistemic status

This treatise makes four kinds of contribution. First, it restates selected results from the three source treatises in a common notation. Second, it supplies a proposed integration architecture. Third, it derives additional propositions under explicit assumptions, including a transaction-conservation result, a finite-program convergence result, a value-of-information result, and a risk-allocation identity. Fourth, it proposes an empirical and institutional research programme.

The distinction between source result and new synthesis matters. The ninefold accounting construction, the AIE program architecture and legitimacy condition, and the multidimensional insurance risk functional are source-derived. The unified state vector, the six-stage Actually protocol, the strengthened convergence assumptions, and the integration theorem are proposed extensions. They should be treated as hypotheses and design principles until tested in formal proofs, institutional pilots, and empirical applications.

2 Conceptual foundations and scope

2.1 What Actually is – and is not

Definition 2.1 (Actually)**.** ACTUALLY is a jointly produced institutional technology that transforms economic observations into verifiable value representations, investigates and resolves material informational asymmetries among affected agents, validates the resulting decisions under ethical and legal constraints, and allocates residual uncertainty through risk-sharing, insurance, or explicit retention.

The definition has five necessary ideas. First, the technology is jointly produced: no party can unilaterally define the common evidentiary standard and then call the result informational symmetry. Second, it is institutional: it requires rules, records, duties, rights, procedures, and means of contestation. Third, it is value-aware: a common information state must describe value trajectories, not merely balances. Fourth, it is normatively constrained: an informed decision is not automatically a legitimate decision. Fifth, it recognises residual uncertainty: what cannot be known or eliminated must be measured, disclosed, and allocated.

Remark 2.2. Actually does not mean total disclosure of every fact to every person. Such a rule would be infeasible, often unlawful, and potentially harmful. The relevant target is the resolution of *material,*

decision-relevant informational differences within a defined programme, subject to privacy, security, confidentiality, and lawful-access constraints. Different agents may retain specialised knowledge while sharing a common verified evidentiary basis.

Remark 2.3. Actually is not synonymous with accounting, auditing, transparency, truth, insurance, or consensus. Each is a component or instrument. A verified record can disclose an unjust transaction; a legitimate objective can be pursued with inadequate accounting; an insurance contract can transfer risk without reducing it; and consensus can be produced by coercion. The integrated system must test each condition separately.

2.2 The unit of analysis

The unit of analysis is a bounded institutional programme: a transaction, contract, enterprise, insurance pool, public project, financial market, cross-border agreement, or other decision system. A programme has a scope, a set of participants, an evidentiary domain, a decision set, and a risk horizon. The system can be nested: a bank-financing decision may contain an enterprise-accounting programme, a collateral-valuation programme, a borrower-disclosure programme, and an insurance programme.

The distinction between local and global completion is important. A bounded programme may terminate even while the wider economy continues to generate new information and new asymmetries. The global ethical system need not terminate for one transaction or institutional relationship to become sufficiently complete for action.

2.3 Design principles

1. **Conservation:** each value-changing transaction must be represented with balanced counterpart entries or an explicitly modelled boundary account.
2. **Traceability:** each material claim must have a source, time, responsible actor, version, and verification status.
3. **Investigability:** every material discrepancy must have an owner, procedure, and disposition.
4. **Legitimacy:** decisions must pass substantive constraints and a defensible process of affected-party consideration.
5. **Risk allocation:** residual uncertainty must be retained knowingly, transferred by contract, pooled, or insured; it must not disappear from the model by omission.
6. **Revisability:** new evidence can reopen a closed programme when a material premise changes or fraud is discovered.
7. **Proportionality:** the cost and intrusiveness of verification must be proportionate to materiality, urgency, reversibility, and potential harm.

3 Primitives, agents, and information states

3.1 Agents and propositions

Let $\mathcal{N} = \{1, \dots, n\}$ be a finite set of agents. Agents may be natural persons, firms, public bodies, insurers, auditors, courts, or states. Let $\mathcal{F}_t$ be the universe of propositions that may be relevant at time t. The information state of agent i is a set $I_i(t) \subseteq \mathcal{F}_t$ containing propositions that the agent treats as available or accepted. The information state may also carry confidence, provenance, and access labels; a simple set representation is used initially to preserve the structure of AIE.

Definition 3.1 (Informational asymmetry)**.** An ordered informational asymmetry between agents i and j at time t is a triple $\alpha = (i, j, \phi)$ such that $\phi \in I_i(t) \setminus I_j(t)$. The total pairwise asymmetry field is

$$\mathcal{A}_\Omega(t) = \bigcup_{i \neq j} \{(i, j, \phi) : \phi \in I_i(t) \setminus I_j(t)\}. \tag{1}$$

Not every difference between information states is material. A doctor and an accountant should not possess identical specialised knowledge. What matters for an institutional decision is whether a proposition could reasonably change valuation, rights, obligations, risk, or the decision of an affected party.

Definition 3.2 (Material asymmetry)**.** For a programme P, let $w_P(\phi)$ measure the materiality of proposition ϕ, and let $\varepsilon_P \geq 0$ be a programme-specific threshold. The material asymmetry field is

$$\mathcal{A}_P^{\mathrm{mat}}(t) = \{(i, j, \phi) \in \mathcal{A}_\Omega(t) : w_P(\phi) \geq \varepsilon_P,\ \phi \text{ is decision-relevant to } P\}. \tag{2}$$

The threshold is not a licence to hide small but cumulative discrepancies. Materiality must be assessed individually and in aggregate. A sequence of individually immaterial errors may become material when their combined effect changes a decision, covenant, solvency assessment, or distributional outcome.

Definition 3.3 (Open material asymmetry)**.** The open material asymmetry set $\mathcal{A}_P^{\mathrm{open}}(t)$ contains those members of $\mathcal{A}_P^{\mathrm{mat}}(t)$ that have neither been resolved through common disclosure and verification nor closed through an authorised, independently verified substitute that makes their decision-relevant implication common to the affected parties. A restricted underlying record may remain confidential if the substitute, restriction, reasons, safeguards, and review rights are adequately recorded.

This is an operational extension of the AIE termination idea. It does not claim that all agents possess identical private information. It requires that no material, decision-relevant difference remain open without a defensible disposition. In the convergence analysis below, A_k denotes this open set at iteration k.

3.2 Evidence objects and provenance

A proposition in an Actually system should be linked to an evidence object. A minimal evidence object is

$$e = (\mathrm{id}, \tau, \mathrm{claim}, \mathrm{source}, \mathrm{actor}, \mathrm{version}, \mathrm{access}, \mathrm{signature}, \mathrm{status}), \tag{3}$$

where τ is the observation or transaction time. The status field distinguishes, at minimum, asserted, corroborated, disputed, rejected, superseded, and legally restricted. A signed or hash-linked record can help prove that an object has not been altered after capture; it does not prove that the original assertion was true. Authenticity of a record and truth of its content are separate predicates.

The provenance layer should preserve the distinction between event time, recording time, verification time, and publication time. This is essential in financial reporting, public procurement, insurance claims, international verification, and forensic review. The W3C provenance model and internal-control frameworks provide relevant technical and governance precedents [49, 16]. Information theory also clarifies that transmission capacity and uncertainty reduction are not the same as semantic truth or justified belief [41].

3.3 Knowledge, belief, and uncertainty

Information and belief must not be conflated. A proposition may be recorded but assigned low confidence; it may be believed but unsupported; or it may be well supported but inaccessible to a particular party. For probabilistic models, let $\pi_i(\theta \mid I_i)$ denote agent i's posterior distribution over an unknown state θ. The distance between beliefs can be measured using a symmetrised divergence, for example

$$D_{ij}(t) = \frac{1}{2}\mathrm{KL}(\pi_i \| m_{ij}) + \frac{1}{2}\mathrm{KL}(\pi_j \| m_{ij}), \qquad m_{ij} = \frac{1}{2}(\pi_i + \pi_j), \tag{4}$$

which is the Jensen–Shannon divergence. This is one possible metric, not a universal measure of ethical importance. Divergence between beliefs can result from different evidence, different models, different priors, or strategic concealment. Investigation must identify which mechanism is present.

4 The value-accounting layer

4.1 The accounting identity as a conservation constraint

Let $A(t)$, $L(t)$, and $E(t)$ denote assets, liabilities, and equity. The accounting identity is

$$A(t) = L(t) + E(t). \tag{5}$$

Equivalently, assets net of liabilities equal equity. The identity constrains the level of a balance sheet but does not, by itself, describe whether the enterprise is appreciating, stagnating, or eroding. This distinction motivates the stochastic decomposition in Value-Based Accounting [2, 23, 24, 32].

4.2 Ninefold trajectory decomposition

Fix a reporting horizon and a materiality threshold. Each asset and liability is classified by the sign of its expected value drift as increasing, constant, or decreasing. The three asset buckets are A_I, A_C, A_D; the three liability buckets are L_I, L_C, L_D. Equity is attributed to corresponding trajectory classes E_I, E_C, E_D. Define the value-state vector

$$X(t) = \big(A_I, A_C, A_D, L_I, L_C, L_D, E_I, E_C, E_D\big)^\top \in \mathbb{R}^9. \tag{6}$$

Let

$$b = \big(1, 1, 1, -1, -1, -1, -1, -1, -1\big)^\top. \tag{7}$$

The admissible value states lie on the hyperplane

$$b^\top X(t) = 0. \tag{8}$$

Thus the nine reported components are subject to one linear balance restriction and occupy an eight-dimensional subspace, before any additional model-specific restrictions are imposed.

4.3 Five operating axioms

The source Value-Based Accounting model is governed by five axioms. They are restated here because they determine what an Actually ledger must preserve.

Axiom 4.1 (Trichotomy)**.** The trajectory classes partition each register:

$$A = A_I + A_C + A_D, \qquad L = L_I + L_C + L_D, \qquad E = E_I + E_C + E_D. \tag{9}$$

Axiom 4.2 (Balance)**.** The value state satisfies $b^\top X(t) = 0$ almost surely at every reporting time.

Axiom 4.3 (Expected trajectory)**.** The expected drifts have the prescribed signs:

$$\nu_{A_I} > 0, \qquad \nu_{A_C} = 0, \qquad \nu_{A_D} < 0, \tag{10}$$
$$\nu_{L_I} > 0, \qquad \nu_{L_C} = 0, \qquad \nu_{L_D} < 0, \tag{11}$$
$$\nu_{E_I} > 0, \qquad \nu_{E_C} = 0, \qquad \nu_{E_D} < 0. \tag{12}$$

Constant value is interpreted in the weak, zero-drift sense unless strict pathwise constancy is explicitly imposed.

Axiom 4.4 (Support and solvency)**.** Asset and liability buckets are supported on non-negative values in the canonical model. Equity buckets may cross zero, thereby recording erosion rather than suppressing it.

Axiom 4.5 (Attribution and purity)**.** Equity value flows are generated only by asset and liability value flows; every primary flow is attributed exactly once according to the canonical equity decomposition.

The canonical equity attribution is

$$dE_I(t) = dA_I(t) - dL_D(t), \tag{13}$$

$$dE_C(t) = dA_C(t) - dL_C(t), \tag{14}$$
$$dE_D(t) = dA_D(t) - dL_I(t). \tag{15}$$

The attribution expresses the economic intuition that asset appreciation and liability amortisation support increasing equity, while asset depreciation and liability accretion erode it. The exact sign convention is inherited from the source model; the classification of a liability is based on the trajectory of the obligation value, not on whether the liability is beneficial to the creditor or debtor.

Proposition 4.6 (Flow conservation)**.** *Under the canonical attribution and the balance identity, the value flows satisfy*

$$dA_I + dA_C + dA_D = dL_I + dL_C + dL_D + dE_I + dE_C + dE_D. \tag{16}$$

After cancellation of the constant-class terms, this becomes

$$dA_I + dA_D = dL_I + dL_D + dE_I + dE_D. \tag{17}$$

Proof. Differentiate the balance identity and use the three-way partitions of assets, liabilities, and equity. The resulting equation is the first displayed relation. Substitute $dE_C = dA_C - dL_C$. The constant-class asset and liability increments cancel, leaving the second relation. This is the pathwise conservation statement established in the source accounting theory [2]. □

4.4 Stochastic dynamics and value momentum

A canonical stochastic specification uses four primitive rates: appreciation α, depreciation β, liability accretion γ, and liability amortisation δ. For Brownian motions $W_1, \ldots, W_6$, one possible model is

$$dA_I = \alpha A_I\, dt + \sigma_1 A_I\, dW_1, \qquad dA_C = \sigma_2 A_C\, dW_2, \tag{18}$$
$$dA_D = -\beta A_D\, dt + \sigma_3 A_D\, dW_3, \qquad dL_I = \gamma L_I\, dt + \sigma_4 L_I\, dW_4, \tag{19}$$
$$dL_C = \sigma_5 L_C\, dW_5, \qquad dL_D = -\delta L_D\, dt + \sigma_6 L_D\, dW_6. \tag{20}$$

The equity processes are then generated by the canonical attribution above. Brownian dependence may be non-zero; independent shocks are not required. The model is a tractable baseline and must be replaced or stress-tested where jumps, heavy tails, regime switching, illiquidity, or strategic reporting dominate.

Let $\nu_Y(t) = \frac{d}{dt}\mathbb{E}[Y(t)]$ denote the expected drift of a value bucket. Define value momentum by

$$M(t) = \nu_{E_I}(t) + \nu_{E_D}(t) = \nu_{A_I}(t) + \nu_{A_D}(t) - \nu_{L_I}(t) - \nu_{L_D}(t). \tag{21}$$

Writing gross value creation and destruction as

$$C(t) = \nu_{A_I}(t) + |\nu_{L_D}(t)|, \tag{22}$$
$$D(t) = |\nu_{A_D}(t)| + \nu_{L_I}(t), \tag{23}$$

we obtain $M(t) = C(t) - D(t)$. Positive momentum indicates that the expected value-creating flows exceed the expected value-destroying flows within the model's scope.

If the expected drift is constant and negative, the arithmetic approximation to the time at which expected equity reaches zero is

$$t^* = \frac{E(0)}{-M}, \qquad M < 0. \tag{24}$$

This is an expected-equity crossing estimate. It is not, without further assumptions, the expected first-passage time of a stochastic equity process. A planning system must distinguish the deterministic forecast crossing from a distribution of insolvency times.

4.5 Transactions as balanced operators

Let a transaction be represented by a vector z_t of transaction quantities, and let N be a matrix mapping transactions into changes in the nine value buckets. Then

$$X_{t+1} = X_t + N z_t. \tag{25}$$

The accounting identity is preserved when

$$b^\top N = 0. \tag{26}$$

Theorem 4.7 (Transaction conservation)**.** *Suppose the initial state is balanced, so that $b^\top X_0 = 0$, and all recorded transaction columns satisfy $b^\top N = 0$. Then every state generated by $X_{t+1} = X_t + N z_t$ is balanced.*

Proof. Assume $b^\top X_t = 0$. Then

$$b^\top X_{t+1} = b^\top (X_t + N z_t) \tag{27}$$
$$= b^\top X_t + (b^\top N) z_t \tag{28}$$
$$= 0 + 0 = 0. \tag{29}$$

Induction from $t = 0$ proves the result for every finite sequence of transactions. □

Remark 4.8. An enterprise is not a closed physical system. Contributions, distributions, taxes, transfers, acquisitions, disposals, and changes in consolidation perimeter must be represented with the appropriate counterpart or boundary account. If an external flow is omitted, a balance residual may reveal the omission; forcing the data to balance by an unexplained plug can conceal it.

Transaction	**Value-flow interpretation**	**Actually control**
Borrowing	Cash and debt rise together; equity need not rise at inception.	Verify lender, principal, maturity, interest, security, covenants, and use of proceeds.
Depreciation or impairment	Asset value moves toward a decreasing trajectory; equity absorbs the attributable loss.	Preserve valuation model, assumptions, approvals, and evidence of impairment.
Debt amortisation	Cash declines and the liability obligation is reduced.	Reconcile payment, lender statement, bank movement, and liability schedule.
Insurance recovery	A covered claim may create a receivable or cash inflow and reduce retained loss.	Verify policy coverage, loss evidence, adjuster decision, deductible, limits, and timing.
Capital contribution	Cash or another asset and contributed equity increase, subject to legal form.	Verify authority, beneficial ownership, consideration, and restrictions.

Table 2: Illustrative transactions and their verification requirements.

4.6 Accounting authenticity is necessary but insufficient

The identity $A = L + E$ can be satisfied by authentic books, erroneous books, fabricated books, or books that omit a material off-balance-sheet obligation. The value-trajectory constraints add dynamic checks but do not prove the truth of the underlying source documents. An actually system therefore separates at least four tests: (i) arithmetic balance; (ii) conservation of recorded flows; (iii) provenance and authenticity of evidence; and (iv) economic plausibility of the trajectory model. The source treatise's ledger-authenticity pipeline, rank checks, and value-health measures provide starting points for these tests [2].

5 The AIE investigation and legitimacy layer

5.1 Ethics programmes

AIE models an ethics programme as a structured process initiated by asymmetric information. For a programme P, write

$$P = \langle E_P, \mathcal{A}_0, \Phi, D_0 \rangle, \tag{30}$$

where E_P is the intended end, $\mathcal{A}_0$ is the initiating asymmetry set, Φ is the investigation function, and D_0 is the set of decisions that investigation may enable. The process evolves as

$$\mathcal{A}_0 \xrightarrow{\Phi} \mathcal{A}_1 \xrightarrow{\Phi} \mathcal{A}_2 \xrightarrow{\Phi} \cdots . \tag{31}$$

An investigation step can reveal previously hidden asymmetries as well as resolve known ones. Let R_k be the asymmetries resolved at step k, and N_k be newly discovered asymmetries. Then

$$\mathcal{A}_{k+1} = (\mathcal{A}_k \setminus R_k) \cup N_k. \tag{32}$$

An expanding asymmetry set is not automatically a failure. It may indicate that the investigation has uncovered hidden dependencies, conflicts of interest, incomplete disclosure, a missing legal authority, or a previously unknown exposure. This dynamic is central to the source AIE account [3].

5.2 Decision enablement and decision closure

AIE distinguishes investigation from decision. A decision becomes actionable only after enough investigation has occurred. For precision, this treatise distinguishes the set of decisions that have become enabled from the set that remains outstanding. Let $\delta(P, t)$ denote the set of decisions enabled by time t, and let $Q_P(t)$ denote the set of decisions still awaiting execution, rejection, or reasoned deferral. For Actually, a programme terminates only when no material asymmetry remains open under the agreed closure standard and no decision remains outstanding:

$$\mathrm{TERM}(P, t^*) \quad \Longleftrightarrow \quad A_P^{\mathrm{open}}(t^*) = \varnothing \quad \wedge \quad Q_P(t^*) = \varnothing. \tag{33}$$

This notation preserves the AIE principle of conjunctive termination while avoiding ambiguity between decisions that have been enabled and decisions that remain uncompleted. A decision is not closed merely because someone has been authorised to make it; the decision must be recorded as executed, rejected, superseded, or deferred under a stated rule and review date.

5.3 The six AIE postulates

The integrated system retains the six structural postulates of AIE, with the termination predicate expressed using the open material asymmetry set defined for Actually:

1. **Compositional structure:** the ethics system is a collection $\Sigma = \{P_1, P_2, \ldots\}$ of programmes with distinct ends, and each programme has an initiating asymmetry set, an investigation function, and a decision set.
2. **Asymmetric initiation:** a programme can be initiated only when a relevant informational asymmetry exists in the total field.
3. **Investigative dynamics:** every active programme applies its investigation function iteratively; an iteration may reduce, expand, or reclassify the open asymmetry set.
4. **Decision enablement:** each decision in the programme's decision set must become actionable through the investigative process before it is decided.
5. **Conjunctive termination:** a bounded programme terminates only when its open material asymmetry set and outstanding decision set are both empty.
6. **Global termination:** the entire ethics system terminates only when every programme in the system terminates.

Global termination is a much stronger condition than local completion. In a changing society, new events and interactions may continuously generate new programmes. Actually therefore certifies bounded institutional programmes rather than claiming that all future ethical inquiry can be completed once and for all.

5.4 A strengthened convergence result

The source AIE treatise identifies finite or bounded asymmetry, investigative honesty, and decisional closure as convergence conditions. For a formal finite-step guarantee, a progress condition is also needed: the investigation must actually remove at least one unresolved asymmetry after the bounded discovery phase. Without this condition, an honest procedure could repeatedly inspect the same unresolved matter without making progress.

Assumption 5.1 (Bounded discovery)**.** There exists a finite iteration K such that $\mathcal{A}_k$ is finite for all $k \leq K$, and no new material asymmetry is introduced after K.

Assumption 5.2 (Investigative honesty)**.** No participant strategically fabricates, suppresses, or reintroduces a material asymmetry that has already been resolved. Material findings are recorded with provenance and are available to the authorised affected parties.

Assumption 5.3 (Progress and fairness)**.** For every $k \geq K$ with $\mathcal{A}_k \neq \varnothing$, the investigation resolves at least one member of $\mathcal{A}_k$ within a finite number of subsequent steps. No unresolved asymmetry is permanently excluded from investigation by strategic scheduling.

Assumption 5.4 (Decisional closure)**.** Every decision that becomes actionable is eventually executed, rejected, or otherwise closed under the programme's lawful decision rule.

Theorem 5.5 (Finite-program convergence)**.** *Under bounded discovery, investigative honesty, progress and fairness, and decisional closure, the open material asymmetry set becomes empty after finitely many investigation steps and the programme terminates, provided that the set of outstanding decisions is finite.*

Proof. At iteration K, the open material asymmetry set $\mathcal{A}_K$ is finite. By bounded discovery, there are no further additions after that point. By investigative honesty, a resolved asymmetry is not strategically reintroduced. By progress and fairness, at least one remaining asymmetry is resolved within a finite number of later steps. Therefore the cardinality of the unresolved set decreases by at least one over each finite block of iterations. A non-negative integer cannot decrease strictly infinitely often. Hence there is a finite iteration K' at which $A_{K'} = \varnothing$. The finite outstanding decision set is closed by decisional closure. Thus both conjuncts in the termination condition hold at a finite time. □

Remark 5.6. The theorem is local to a bounded programme. It does not imply that a society, market, or global economy will permanently exhaust every informational asymmetry. New agents, technologies, shocks, laws, and preferences can initiate new programmes. AIE's broader observation that a global ethics system may be non-terminating is therefore compatible with finite termination of a particular Actually programme [3].

5.5 Legitimacy of the end

The AIE source introduces three conditions for the legitimacy of an end. Let $C = \{c_1, \ldots, c_m\}$ be a set of peremptory constraints, and let $V(a)$ be the revealed-value set of initiating agent a. Denote the programme end by E_P, to distinguish it from accounting equity $E(t)$. The end is admissible only if:

$$V_1(E_P): \quad E_P \text{ does not require violation of any } c_j \in C, \tag{34}$$

$$V_2(E_P): \quad E_P \in V(a), \tag{35}$$

$$V_3(E_P): \quad \text{every affected agent would endorse } E_P \text{ under the specified full-information condition.} \tag{36}$$

Definition 5.7 (Legitimate end)**.** An end E_P is legitimate for programme P if and only if $V_1(E_P) \wedge V_2(E_P) \wedge V_3(E_P)$.

Proposition 5.8 (Legitimacy is conjunctive)**.** *If any one of the conditions V_1, V_2, V_3 fails, the end is not legitimate under the AIE criterion.*

Proof. This follows directly from the conjunction in the definition. A preference consistently pursued by an agent cannot override a peremptory constraint; compliance with a constraint does not establish that the end is genuinely valued by the initiator; and an initiator's revealed preference does not establish that the end can be justified to affected parties under conditions of full disclosure. Each condition addresses a distinct failure mode [3, 21, 37]. □

The ideal-endorsement condition is a normative criterion, not a claim that unanimity can be mechanically measured. Real institutions need procedures for representation, accessibility, dissent, reason-giving, appeal, and protection against coercion. AIE's conflict hierarchy of Pareto improvement, weighted priority, and adjudication can guide conflicts among programmes, but weighting must never override non-derogable rights or lawful constraints [3, 35].

6 The insurance and residual-risk layer

6.1 Why information cannot eliminate risk

Even a fully documented and honestly disclosed transaction remains exposed to events that are not known in advance. Insurance is therefore not a substitute for disclosure; it is the mechanism for dealing with risk that persists after reasonable investigation. Classical insurance often begins with expected loss and a loading. The general insurance source broadens this to physical, actuarial, systemic, model, and aggregate uncertainty [4, 8, 11, 9].

Let L_i be the loss associated with insured exposure i, and let aggregate loss be

$$L_T = \sum_{i=1}^{n} L_i. \tag{37}$$

The source insurance framework distinguishes the following components:

$$A_i^{\text{phys}} = \mathbb{E}[\text{Var}(L_i \mid \theta_i)], \tag{38}$$

$$U_i^{\text{act}} = \text{Var}(\mathbb{E}[L_i \mid \theta_i]), \tag{39}$$

$$S_i^{\text{sys}} = \text{Cov}(L_i, L_T), \tag{40}$$

$$M_i^{\text{model}} = \text{Var}\left(\mathbb{E}[L_i \mid \mathcal{M}^{(k)}]\right), \tag{41}$$

where θ_i denotes loss parameters and $\mathcal{M}^{(k)}$ denotes a family of competing models. A candidate individual uncertainty score is

$$R_i = \alpha A_i^{\text{phys}} + \beta U_i^{\text{act}} + \gamma S_i^{\text{sys}} + \delta M_i^{\text{model}}. \tag{42}$$

The aggregate functional may be written as

$$R_T = \rho(L_T) + \eta\Phi + \zeta\Psi, \tag{43}$$

where ρ is an aggregate risk measure and Φ, Ψ represent parameter and model uncertainty at the aggregate level. In implementation, terms must be normalised or expressed in compatible units; otherwise their weighted sum has no coherent financial interpretation.

6.2 Bayesian updating of insured risk

Let D_i be the evidence received about exposure i, and let $\pi_i(\theta_i)$ be the prior distribution over its loss parameters. Bayes' rule gives

$$\pi_i(\theta_i \mid D_i) = \frac{p(D_i \mid \theta_i)\pi_i(\theta_i)}{\int p(D_i \mid \vartheta)\pi_i(\vartheta)\, d\vartheta}. \tag{44}$$

The posterior expected loss and parameter uncertainty are

$$\mathbb{E}[L_i \mid D_i] = \int \mathbb{E}[L_i \mid \theta_i]\pi_i(\theta_i \mid D_i)\, d\theta_i, \tag{45}$$

$$U_i^{\text{act}}(D_i) = \text{Var}_{\pi_i(\theta_i \mid D_i)}\left(\mathbb{E}[L_i \mid \theta_i]\right). \tag{46}$$

This makes the information channel explicit. Better data can narrow the posterior distribution, but the posterior mean loss can rise or fall. An actually process therefore reports both the updated expected claim and the updated uncertainty measure.

6.3 Risk measurement and risk allocation

A risk measure maps a random loss exposure into a capital or risk quantity. Coherent risk measures satisfy monotonicity, translation invariance, positive homogeneity, and subadditivity under the usual sign conventions [9]. Subadditivity captures a diversification property, but it does not imply that every pool is safer in every scenario. Strong dependence, common shocks, concentration, counterparty failure, or model error can remove diversification benefits.

Proposition 6.1 (Subadditive pooling)**.** *For a coherent risk measure ρ, the risk of a pooled portfolio is no greater than the sum of the stand-alone risks:*

$$\rho\left(\sum_{i=1}^{n} Z_i\right) \leq \sum_{i=1}^{n} \rho(Z_i). \tag{47}$$

Proof. The result follows by repeated application of subadditivity, one of the defining properties of a coherent risk measure. It does not imply that every member pays less after pooling, because allocation, expenses, moral hazard, capital costs, and distributional rules remain relevant. □

Assumption 6.2 (Differentiable homogeneous risk measure)**.** Let $Z_1, \ldots, Z_n$ be random loss components, let $x_i \geq 0$ be exposure scales, and define $g(x) = \rho(\sum_i x_i Z_i)$. Assume that g is differentiable and positively homogeneous of degree one, so that $g(cx) = cg(x)$ for every $c > 0$.

Theorem 6.3 (Euler allocation of residual risk)**.** *Under the differentiability and positive-homogeneity assumption, the aggregate risk capital can be allocated as*

$$c_i = x_i \frac{\partial g(x)}{\partial x_i}, \tag{48}$$

and the allocations satisfy

$$\sum_{i=1}^{n} c_i = g(x) = \rho\left(\sum_{i=1}^{n} x_i Z_i\right). \tag{49}$$

Proof. Euler's theorem for differentiable homogeneous functions of degree one states that

$$g(x) = \sum_{i=1}^{n} x_i \frac{\partial g(x)}{\partial x_i}. \tag{50}$$

Defining each term on the right as c_i proves the allocation identity. The result allocates the risk measure; it does not, by itself, specify the entire premium, which must also account for expected claims, expenses, taxes, reinsurance, capital costs, and contract terms. □

A premium schedule consistent with this decomposition can be written as

$$P_i = \mathbb{E}[I_i] + \lambda c_i + e_i, \tag{51}$$

where I_i is the indemnity, λ converts risk capital into a monetary loading, and e_i is the allocated expense and other explicitly specified loadings. Under consistent allocation,

$$\sum_{i=1}^{n} P_i = \mathbb{E}\left[\sum_{i=1}^{n} I_i\right] + \lambda g(x) + \sum_{i=1}^{n} e_i. \tag{52}$$

For non-differentiable risk measures, a subgradient or other allocation rule is needed. Cooperative-game allocations, including the Shapley value, provide an alternative when coalition contributions can be estimated [42]. For a cooperative risk game with coalition cost $v(S)$, the Shapley allocation to participant i is

$$\phi_i(v) = \sum_{S \subseteq \mathcal{N} \setminus \{i\}} \frac{|S|!(n - |S| - 1)!}{n!} \left[v(S \cup \{i\}) - v(S)\right]. \tag{53}$$

This allocates the average marginal contribution across all coalition orderings. It can support a fairness analysis, but its result depends on the coalition definition, the chosen cost function, transferable-utility assumptions, and the quality of the loss data.

Remark 6.4. The source General Theory of Insurance motivates premiums by marginal contributions to a global uncertainty functional. The Euler construction above supplies a more explicit allocation result under differentiability and positive homogeneity. It should not be read as claiming that every derivative of every uncertainty functional is automatically a monetary premium.

6.4 The value of better information

Let $\mathcal{F} \subseteq \mathcal{G}$ be two information sets, with $\mathcal{G}$ at least as informative as $\mathcal{F}$. Let $\ell(\theta, d)$ be a decision loss and let $\mathcal{D}(\mathcal{H})$ be the set of decision rules measurable with respect to information set $\mathcal{H}$. Define ex ante Bayes risk by

$$\mathcal{B}(\mathcal{H}) = \inf_{\delta \in \mathcal{D}(\mathcal{H})} \mathbb{E}[\ell(\theta, \delta)]. \tag{54}$$

Proposition 6.5 (More information weakly reduces optimal decision risk)**.** *If the decision-maker can use any decision rule available under $\mathcal{F}$ after observing $\mathcal{G}$, and the objective and action set are unchanged, then ex ante Bayes risk under $\mathcal{G}$ is no greater than under $\mathcal{F}$.*

Proof. Every decision rule measurable with respect to $\mathcal{F}$ is also measurable with respect to $\mathcal{G}$. The decision-maker observing $\mathcal{G}$ can therefore ignore the additional signal and implement the optimal $\mathcal{F}$-based rule. Optimisation over the larger set of admissible rules cannot yield a higher minimum expected loss. □

This is a decision-theoretic value-of-information result consistent with Blackwell's comparison of experiments [10]. It does not imply that better information always lowers an insurance premium. Better information may reveal that expected claims are higher than previously believed, alter risk classification, or expose concentration. It can reduce estimation uncertainty while increasing the measured expected loss. Actually requires that these effects be separated rather than conflated.

6.5 Insurance governance

An insurance contract is itself an information and incentive structure. The insured knows some features of its exposure better than the insurer; the insurer knows some features of its pricing model and claims practice better than the insured. Underwriting, exclusions, deductibles, claims adjudication, reserving, reinsurance, and capital rules can all generate new material asymmetries. The insurance layer must therefore be governed by the same Actually process: disclose the relevant policy terms, explain model limitations, preserve claims evidence, provide review and appeal, and record the residual risk retained by each party.

7 The unified Actually architecture

7.1 System state

At time t, define the integrated state

$$Z_t = (X_t, \mathcal{I}_t, A_t, Q_t, \mathcal{E}_t, \mathcal{R}_t, \mathcal{K}_t, \mathcal{G}_t), \tag{55}$$

where X_t is the value-state vector; $\mathcal{I}_t$ is the collection of agent information states; A_t is the material asymmetry set; Q_t is the outstanding decision set; $\mathcal{E}_t$ is the evidence and provenance register; $\mathcal{R}_t$ is the residual-risk model; $\mathcal{K}_t$ is allocated risk capital, insurance, and retained exposure; and $\mathcal{G}_t$ is the governance and legal-authority record.

An Actually operation maps an observed event and the current institutional state into a revised state and a decision certificate:

$$\textsc{Actually} : (Z_t, \text{event}_{t+1}) \longmapsto (Z_{t+1}, \text{certificate}_{t+1}). \tag{56}$$

The certificate is not a metaphysical proof that every claim is true. It is a structured record showing which claims were observed, how they were verified, which discrepancies were resolved, which decisions were taken, what risk remains, and who is responsible for it.

7.2 Six-stage protocol

1. **Observation and registration.** Capture events, transactions, claims, and obligations with time, source, actor, and version metadata.

2. **Valuation and conservation.** Map transactions into the nine value buckets, verify balance and conservation, and estimate value trajectories.

3. **Asymmetry detection.** Compare material, decision-relevant information states and identify discrepancies in facts, assumptions, access, authority, or interpretation.

4. **Investigation and legitimacy.** Apply the investigation function, record newly discovered asymmetries, enable decisions only after sufficient review, and test proposed ends against the legitimacy conditions.

5. **Residual-risk allocation.** Quantify physical, actuarial, systemic, model, legal, operational, and geopolitical risks; allocate them by insurance, contractual retention, reserves, or other consent-based mechanisms.

6. **Execution, feedback, and reopening.** Execute or close decisions, settle obligations, update the ledger and risk model, and reopen a programme when new material evidence or a qualifying event occurs.

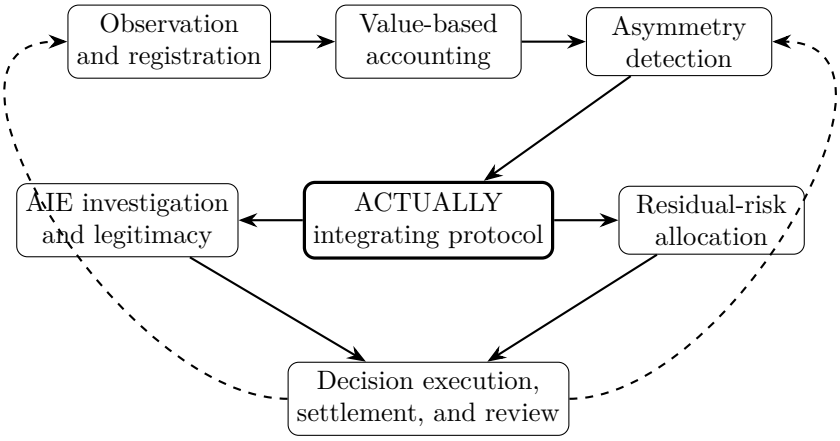


Figure 1: The integrated Actually architecture. Accounting, AIE, and insurance remain distinct technical layers coordinated by a common institutional protocol.

7.3 Actually completion conditions

Definition 7.1 (Actually-complete programme)**.** A programme P is Actually-complete at time t^* if and only if all of the following predicates hold:

$$\mathrm{Bal}(P,t^*): \quad b^\top X_{t^*} = 0 \text{ and all relevant transaction flows are reconciled,} \tag{57}$$

$$\mathrm{Prov}(P,t^*): \quad material evidence has adequate provenance and its verification status is recorded, \tag{58}$$

$$\mathrm{Conv}(P,t^*): \quad A_P^{\mathrm{open}}(t^*) = \varnothing \text{ within the agreed scope,} \tag{59}$$

$$\mathrm{Dec}(P,t^*): \quad Q_P(t^*) = \varnothing, \tag{60}$$

$$\mathrm{Leg}(P,t^*): \quad \text{the proposed end satisfies the applicable legitimacy conditions,} \tag{61}$$

$$\mathrm{Risk}(P,t^*): \quad \text{residual risk is disclosed, allocated, and financed or explicitly retained,} \tag{62}$$

$$\mathrm{Law}(P,t^*): \quad \text{the process respects applicable law, rights, authority, and review procedures.} \tag{63}$$

Theorem 7.2 (Actually soundness under stated assumptions)**.** *Suppose a bounded programme satisfies transaction conservation, evidence-provenance validation, finite-program convergence, decision closure, legitimacy, lawful authority, and a solvent risk-allocation arrangement. Then its completion certificate establishes (i) accounting consistency of the represented value state, (ii) closure or authorised disposition of all open material informational asymmetries within the programme's declared scope, (iii) closure of the programme's decisions, (iv) legitimacy of the authorised end under the adopted criterion, and (v) explicit allocation or retention of modelled residual risk.*

Proof. Transaction conservation and the initial balance imply the accounting identity at every recorded step by the transaction-conservation theorem. Provenance validation establishes the stated chain of custody for the evidence objects; it does not prove the truth of evidence beyond the validation standard. Finite-program convergence and decision closure imply that the open material-asymmetry and outstanding-decision sets are empty within the declared scope. The legitimacy assumption supplies the normative admissibility of the end. Finally, the risk-allocation arrangement identifies the insured, transferred, reserved, and retained components of the modelled residual risk and satisfies its stated solvency condition. These conclusions are exactly the predicates recorded in the completion certificate. The theorem is conditional: an omitted event, false source document, invalid legal authority, or misspecified risk model can invalidate its premises. □

7.4 A certificate as a portable institutional object

A certificate should contain at least: programme identifier and scope; parties and authorised representatives; accounting period and value-state snapshot; transaction reconciliation; evidence register and provenance references; material asymmetry register; investigation history; decision log and dissent record; legitimacy review; insurance and retained-risk schedule; legal basis and appeal route; model versions; and reopening triggers. A cryptographic digest can bind the certificate to the underlying records, while access controls determine which parties may inspect protected information. A digest proves a relationship to a particular record; it does not prove that the record is factually accurate.

8 Financial planning, budgeting, and analysis

8.1 From ledger to forward plan

Financial planning and analysis (FP&A) is a natural application because it joins accounting states, operating assumptions, decisions, and risk. An Actually-enabled forecast must make explicit the source and owner of each material assumption. For an enterprise, a minimal forecast includes revenue, operating expenses, working capital, capital expenditure, debt service, tax, insurance recoveries, and equity. A forecast without traceable assumptions is a scenario narrative rather than an auditable plan.

Let B_t be budgeted performance and Y_t realised performance. The variance is

$$\Delta_t = Y_t - B_t. \tag{64}$$

Every material variance should be classified as a volume, price, mix, timing, estimation, model, or external-shock effect. The classification is itself an investigative claim and can be disputed. Actually therefore requires a variance bridge with linked source records and a decision owner.

8.2 Liquidity, solvency, and capital allocation

Let cash at time t be C_t, and let average net cash burn over a relevant horizon be $b_t > 0$. A simple liquidity-runway measure is

$$\text{Runway}_t = \frac{C_t}{b_t}. \tag{65}$$

It must be stress-tested for seasonality, committed facilities, restricted cash, refinancing risk, and contingent obligations. A firm can report positive accounting equity and still fail to meet near-term payments; solvency and liquidity are therefore distinct.

The value-momentum measure M_t can be included in management reporting alongside cash runway, debt-service coverage, covenant headroom, expected credit loss, and risk capital. A negative M_t is a warning about expected value erosion; it is not a substitute for cash-flow forecasting or legal insolvency tests.

A capital-planning decision should compare projects under a common information and risk basis. For project j, let cash flows be $CF_{j,t}$, discount factors be d_t, and risk-adjustment be K_j. A simplified decision score is

$$V_j = \sum_{t=0}^{T} d_t \mathbb{E}[CF_{j,t}] - K_j. \tag{66}$$

Actually requires the expected cash flows, discount factors, risk adjustment, and distributional effects to have traceable provenance. The decision should also disclose non-financial constraints, dependencies, and downside scenarios. A positive net present value does not make an unlawful, coercive, or illegitimate project acceptable.

8.3 Scenario analysis and stress testing

A robust FP&A process includes a base case, adverse case, severe-but-plausible case, and reverse stress test. Scenarios should vary demand, input costs, interest rates, exchange rates, counterparty default, insurance recoveries, cyber incidents, physical shocks, and geopolitical disruptions. The model should identify which assumptions are observable, estimated, judgmental, or imposed by policy. A scenario's value lies not in its precision but in making conditional consequences comparable.

Planning object	**Core measure**	**Actually question**
Liquidity	Cash runway and committed liquidity	Are cash balances unrestricted, and are facilities legally available under stress?
Profitability	Operating margin and value momentum	Are gains recurring, attributable, and supported by source transactions?
Solvency	Equity, leverage, and debt-service coverage	Are obligations complete, correctly valued, and mapped to maturities?
Capital expenditure	NPV, payback, and scenario range	Are costs, benefits, discount rates, and project dependencies independently reviewable?
Insurance	Premium, limits, exclusions, and retained loss	Which losses remain uninsured or subject to disputed interpretation?

Table 3: An Actually-enabled FP&A dashboard.

9 Statistics, econometrics, and empirical identification

9.1 State-space representation

The ninefold value state is constrained by the balance equation. Let $H \in \mathbb{R}^{9\times 8}$ be a full-column-rank matrix whose columns span the null space of $b^\top$, so that

$$b^\top H = 0. \tag{67}$$

Every balanced state can be represented as $X_t = Hz_t$, where $z_t \in \mathbb{R}^8$. A state-space model is

$$z_t = F_t z_{t-1} + G_t u_t + w_t, \tag{68}$$
$$y_t = C_t H z_t + D_t a_t + \epsilon_t, \tag{69}$$

where y_t contains observed disclosures, u_t contains known control inputs, w_t is state noise, a_t captures unobserved or disputed informational effects, and ϵ_t is measurement error. The distinction between a_t and ϵ_t is important: missing information is not necessarily random measurement noise. Kalman filtering provides the linear state-estimation benchmark, while broader time-series methods are required for nonstationarity and regime change [26, 22].

For a linear Gaussian benchmark, the Kalman prediction step is

$$\widehat{z}_{t|t-1} = F_t \widehat{z}_{t-1|t-1} + G_t u_t, \tag{70}$$
$$P_{t|t-1} = F_t P_{t-1|t-1} F_t^\top + Q_t. \tag{71}$$

The innovation and update are

$$e_t = y_t - C_t H \widehat{z}_{t|t-1}, \tag{72}$$

$$S_t = C_t H P_{t|t-1} H^\top C_t^\top + R_t, \tag{73}$$
$$K_t = P_{t|t-1} H^\top C_t^\top S_t^{-1}, \tag{74}$$
$$\widehat{z}_{t|t} = \widehat{z}_{t|t-1} + K_t e_t. \tag{75}$$

The measurement equation can be extended to robust, non-Gaussian, jump, or regime-switching models. A filter cannot solve identification failures by computation alone. Independent anchors, repeated observations, structural restrictions, and adversarial validation are required.

9.2 Ledger and disclosure diagnostics

A practical empirical system should estimate at least the following diagnostics:

$$\text{BalanceResidual}_t = b^\top X_t, \tag{76}$$
$$\text{FlowResidual}_t = \Delta A_{I,t} + \Delta A_{D,t} - \Delta L_{I,t} - \Delta L_{D,t} - \Delta E_{I,t} - \Delta E_{D,t}, \tag{77}$$
$$\text{DisclosureLag}_i = t_{\text{verified},i} - t_{\text{observed},i}, \tag{78}$$
$$\text{Coverage}_P = \frac{\sum_{q \in \mathcal{Q}_P^{\text{verified}}} w_q}{\sum_{q \in \mathcal{Q}_P^{\text{material}}} w_q}. \tag{79}$$

A high verification coverage score does not prove that the omitted remainder is unimportant. The denominator and weights must be independently reviewed, and sensitivity analysis should test whether the conclusion changes under alternative materiality weights.

9.3 Causal evaluation of Actually adoption

Suppose firms or institutions adopt an Actually protocol at different times. Let Y_{it} be an outcome such as audit restatements, financing costs, claims disputes, payment delays, or forecast errors. A two-way fixed-effects starting specification is

$$Y_{it} = \alpha_i + \lambda_t + \beta \, \text{Adopt}_{it} + \gamma^\top W_{it} + \varepsilon_{it}, \tag{80}$$

where α_i are unit effects, λ_t are time effects, and W_{it} are time-varying covariates. The coefficient β is not automatically causal. The evaluation must assess parallel trends, anticipation, treatment heterogeneity, spillovers, selection into adoption, measurement changes, and concurrent reforms. Event-study designs, matched comparisons, synthetic controls, and randomised phased pilots may be appropriate depending on the adoption process [6, 34].

9.4 Testable hypotheses

1. **H1 – Reconciliation:** adoption reduces unexplained accounting and transaction-flow residuals, conditional on stable scope and measurement.
2. **H2 – Disclosure quality:** adoption reduces the time between observation, verification, and authorised disclosure for material claims.
3. **H3 – Dispute resolution:** adoption reduces the incidence or duration of disputes caused by preventable information gaps, after controlling for transaction complexity.
4. **H4 – Risk pricing:** adoption reduces the uncertainty component of pricing when additional evidence is informative, while the expected-loss component may rise or fall.
5. **H5 – Distribution:** adoption improves the alignment between risk-bearing capacity, contractual responsibility, and actual loss incidence, without systematically shifting costs to less powerful parties.
6. **H6 – Resilience:** adoption improves stress-test readiness and recovery time following material shocks, conditional on adequate capital and institutional capacity.

10 Law, contracts, and institutional governance

10.1 The legal function

Law gives economic information rights and duties an enforceable institutional form. Depending on the context, relevant doctrines may include accounting and disclosure obligations, fiduciary duties, contract interpretation, agency, consumer protection, privacy, data protection, insurance regulation, insolvency, competition, administrative law, and evidence. Actually does not replace any of them. It supplies a framework for identifying which legal duties govern a particular programme and for recording compliance, dispute, and remedy.

A legal governance record should identify: parties and legal capacity; authorisations and delegations; governing law and jurisdiction; disclosure duties; confidentiality and data-access rules; retention and deletion rules; standards of evidence; conflicts of interest; appeal and adjudication routes; and remedies for misrepresentation or non-performance. In cross-border settings, applicable treaty obligations and conflict-of-law rules must be separately examined. The Vienna Convention on the Law of Treaties and the United Nations Charter are foundational legal instruments, but neither automatically validates a private accounting or insurance arrangement [48, 47].

10.2 Disclosure, privacy, and due process

Informational symmetry is not a warrant for unrestricted surveillance. Personal, medical, commercial, and security-sensitive information may require confidentiality, purpose limitation, access controls, and minimisation. Actually should disclose to each party the material information needed for a legitimate decision while protecting information that is not lawfully or reasonably shareable. Where full disclosure is prohibited, the programme should record the restriction and use authorised substitutes such as independent review, redaction, escrow, or confidential adjudication.

A decision process should preserve notice, opportunity to respond, access to reasons, impartial review, and proportional remedies. The stronger the potential harm and irreversibility, the stronger the required procedural safeguards. This is particularly important where a participant has less bargaining power or depends on an institution for essential services.

10.3 Institutional separation of powers

No single actor should ordinarily control the entire chain of assertion, valuation, verification, adjudication, and payment. Separation reduces conflicts of interest. Institutional analysis of collective action also emphasises the importance of rules-in-use, monitoring, graduated sanctions, and accessible conflict-resolution mechanisms [33]. A practical structure distinguishes: (i) originating party; (ii) record keeper; (iii) independent verifier; (iv) decision authority; (v) risk bearer or insurer; and (vi) appeal body. In small programmes where functional separation is impossible, compensating controls and explicit conflict declarations are required.

11 Physics, chemistry, biology, and systems theory

11.1 Physics: conservation and symmetry

The accounting identity resembles a conservation constraint: changes in one register must be matched by corresponding changes elsewhere. In mathematical physics, Noether's theorem links continuous symmetries of an action to conserved quantities under specified regularity and dynamical assumptions [29]. The analogy is useful but must not be overstated. The accounting identity is imposed by the structure of the ledger and the chosen boundary; it is not automatically a Noether charge derived from a physical Lagrangian. A Noether interpretation would require a genuine dynamical model with an appropriate symmetry.

The practical lesson is that a change of representation should not create or destroy the underlying value claim. Currency denomination, valuation basis, consolidation method, or numeraire can alter displayed coordinates while leaving the properly transformed economic relation invariant. This suggests testing

both invariance of the accounting identity and sensitivity of trajectory classifications to measurement conventions.

11.2 Chemistry: stoichiometric networks

A reaction network can be represented by a stoichiometric matrix S and a vector of reaction rates $v(x)$:

$$\dot{x} = Sv(x). \tag{81}$$

If a vector ℓ satisfies $\ell^\top S = 0$, then

$$\frac{d}{dt}(\ell^\top x) = \ell^\top S v(x) = 0. \tag{82}$$

The conserved moiety is analogous to a ledger invariant. The correspondence is methodological: both systems identify admissible transformations and conserved quantities. Economic value is not a chemical substance, and accounting classes are not chemical species. The useful borrowing is the use of network structure and invariants to detect inconsistent flows.

11.3 Biology: homeostasis and feedback

Living systems maintain selected state variables within viable regions through feedback, adaptation, and resource allocation. A stylised local model is

$$\dot{h} = f(h, u, d), \tag{83}$$

where h is a health state, u is a control response, and d is a disturbance. Actually uses a comparable feedback architecture: new observations update the state estimate; detected discrepancies trigger investigation; and residual risks trigger reserves, insurance, or adaptation. The biological analogy concerns feedback and resilience, not a claim that an economy is literally an organism. Economic institutions also contain conflict, agency, power, and rights that biological systems do not encode in the same way [13, 39].

12 Linguistics, psychology, psychiatry, and philosophy

12.1 Linguistics: shared meaning and translation

Disclosure is not complete merely because the same words have been transmitted. Parties must share enough context to interpret the claims in a compatible way. Pragmatics and theories of common ground distinguish the information that is publicly accepted in a conversation from the private beliefs held by individual speakers [44, 20, 14]. Actually should therefore preserve the language of an original claim, its translation, the translator or translation model, contested terms, and the authoritative version for decision purposes.

For multilingual contracts, financial statements, treaties, and technical standards, a glossary of defined terms is part of the control environment. A translation can preserve literal wording while changing legal or economic implication. The system should use bilingual or multilingual review for high-materiality terms, maintain version control, and record disagreements rather than silently harmonising them.

12.2 Psychology: bounded judgement and trust

Agents do not process all evidence as ideal Bayesian statisticians. Attention is scarce; framing, loss aversion, overconfidence, anchoring, motivated reasoning, and group identity can affect what is noticed and how it is interpreted [46, 25]. Actually should not presume that a disclosure has been understood merely because it was delivered. Material disclosures may require comprehension checks, accessible formats, cooling-off periods, independent advice, and decision aids.

Trust is neither blind confidence nor universal suspicion. It is a warranted expectation built from verifiable conduct, institutional competence, and remedies for failure. A credible actually system makes it possible to challenge a claim without treating every disagreement as fraud.

12.3 Psychiatry and health information

Psychiatric and medical records are especially sensitive because disclosure can create stigma, discrimination, or coercion. A patient, clinician, insurer, employer, and regulator do not need identical information states. They need a lawful and proportionate arrangement in which the information required for care, consent, coverage, or oversight is available to the relevant authorised party. Actually is not a psychiatric diagnostic tool and should never classify a person as dishonest, irrational, or dangerous from a statistical anomaly. Health applications require clinical governance, informed consent, privacy safeguards, and human professional judgement.

12.4 Philosophy: truth, justification, and legitimacy

Actually distinguishes three philosophical questions:

1. **Truth:** is the proposition accurate about the relevant world?
2. **Justification:** is there adequate evidence and a defensible inference supporting the proposition?
3. **Legitimacy:** is the proposed action justified to those affected, consistent with binding constraints, and adopted through a fair process?

A claim can be true but unjustified by the evidence available to a decision-maker; a belief can be justified but false; and a technically well-supported decision can still be illegitimate because it violates rights or imposes unjustifiable harm. This separation connects epistemology to public reason, discourse ethics, and contractualist justification [21, 37, 35].

13 Religion, welfare, and political economy

13.1 Religious and ethical institutions as value-flow arrangements

The Value-Based Accounting source identifies Jubilee, limits on usury, and zakat among religiously grounded institutions that can be interpreted as mechanisms governing value flows, debt, and redistribution [2]. The present treatise treats these as comparative institutional examples, not as doctrinally identical systems. Their legal status, theological interpretation, historical implementation, and distributive effects differ across traditions and jurisdictions.

The general question is whether institutions can prevent value claims from being concealed, concentrated without justification, or transferred onto parties who cannot reasonably understand or contest them. Religious and secular institutions alike can contribute principles of stewardship, charity, debt relief, fair dealing, solidarity, restraint, and responsibility. Their operation must still be tested for transparency, accountability, rights, and actual welfare effects.

13.2 Welfare economics and distribution

Insurance can improve welfare by reducing consumption volatility and pooling idiosyncratic losses. But risk pooling may be regressive if premiums are unaffordable, exclusions are opaque, or correlated risks are transferred to parties least able to bear them. Let household consumption be c_i and welfare weight be ω_i. A planner may consider

$$W = \sum_{i=1}^{n} \omega_i u_i(c_i), \tag{84}$$

subject to resource, incentive, solvency, legal, and participation constraints. The weights are normative choices and must be disclosed. A value-neutral risk score cannot determine the social welfare weights.

An Actually evaluation should report not only aggregate surplus but also who receives gains, who pays premiums, who retains uncovered risk, whose data are used, and who has access to appeal. Distributional accounting is necessary because an aggregate improvement can coexist with serious losses for a subgroup. Development economics and capability approaches emphasise that welfare must be assessed in terms of substantive opportunities as well as monetary output [40].

13.3 Political science and institutional credibility

Public institutions produce rules, records, budgets, and commitments. Their credibility depends on administrative capacity, independent oversight, predictable enforcement, and public reasons. A government that publishes budgets but does not reconcile commitments, contingent liabilities, procurement obligations, or public guarantees has not achieved complete fiscal transparency. Conversely, disclosure without privacy and security safeguards can expose citizens or critical infrastructure.

Performance indicators can become targets and be manipulated. Goodhart-type effects mean that the system must retain qualitative review and audit of the indicator itself [19, 12]. Actually therefore includes meta-verification: who selected the metric, what it omits, how it can be gamed, and what happens when the metric conflicts with lived outcomes.

14 Geopolitics, diplomacy, international relations, and warfare economics

14.1 Cross-border informational asymmetry

States, central banks, development banks, firms, insurers, and multilateral institutions often operate with different legal mandates, accounting conventions, languages, data access, and political incentives. A cross-border Actually programme would specify common definitions, reporting units, valuation dates, audit access, dispute procedures, and rules for confidential information. It should distinguish a statistical disagreement from a legal disagreement and both from a political disagreement.

Diplomatic verification is one institutional analogue. A treaty obligation becomes more credible when the parties agree on observable indicators, inspection rights, reporting schedules, procedures for disputed evidence, and proportionate remedies. The legal principle of pacta sunt servanda and the rules governing treaty interpretation provide a normative framework, but the design of a verification mechanism remains context-specific [48, 47].

14.2 Development finance and sovereign risk

Sovereign financing combines fiscal capacity, foreign-currency obligations, contingent liabilities, institutional credibility, external shocks, and political risk. Actually can be applied to public debt registers, guarantees, public-private partnerships, infrastructure finance, and disaster-risk pools. It requires consolidated reporting across budgetary and off-budget commitments, stress-testing of foreign-exchange and interest-rate exposures, and explicit allocation of construction, demand, political, and climate risks.

No single credit rating or debt ratio can replace this analysis. A sovereign may be solvent under one growth, interest-rate, and exchange-rate path but vulnerable under another. The relevant question is not whether risk has a single true number; it is whether the assumptions, scenarios, evidence, and decision rights are visible and contestable.

14.3 Warfare economics and civilian protection

War is a regime of correlated losses: supply chains fail together, infrastructure is damaged, insurance capacity can withdraw, payment channels may be disrupted, and model uncertainty rises. A general insurance system must therefore allow for concentration, exclusions, counterparty default, and the possibility that a risk is uninsurable at ordinary prices. Emergency financing and public guarantees can shift rather than eliminate these risks.

Actually's role in warfare economics is defensive and institutional: maintain auditable records of public resources, protect civilian access to essential services, identify supply-chain dependencies, make humanitarian expenditure traceable, and disclose the distribution of losses. It does not authorise attacks or make operational military decisions. Any application in armed conflict must remain subordinate to international humanitarian law, including distinction, proportionality, precautions, humane treatment, and protection of civilians and civilian objects [18, 7, 15, 38]. The strategic literature is relevant to understanding coercion and escalation; it is not a substitute for legal obligations.

15 Computer science, algorithms, data science, and AI/ML

15.1 System architecture

A software implementation can be separated into modules: event ingestion; identity and authority management; ledger and valuation; provenance; asymmetry detection; investigation workflow; decision and consent records; risk estimation; insurance allocation; audit; and reporting. Each module should expose versioned interfaces and machine-readable records. The platform should support export and independent verification so that no vendor or central administrator becomes the sole holder of institutional truth.

15.2 Core algorithm

The following pseudocode specifies the control flow. It is intentionally expressed without a specialised algorithm package.

```
PROCEDURE ACTUALIZE(program P, event stream E):
    INITIALISE value_state X, evidence_register R,
               asymmetry_set A, pending_decisions Q,
               residual_risk model U, governance_record G

    FOR each event e in E:
        authenticate source and record provenance(e, R)
        classify event and map it to transaction vector z
        verify accounting conservation for z
        update X using the balanced transaction operator
        estimate value trajectories and value momentum

        detect material information differences among authorised parties
        add newly discovered asymmetries to A

        WHILE material asymmetries remain in scope:
            assign each asymmetry an owner, evidence standard, and deadline
            investigate; preserve new findings and dissent
            resolve verified discrepancies or record reasoned exceptions
            enable only those decisions supported by sufficient inquiry
            add newly enabled but uncompleted decisions to Q

        test proposed ends against legal and ethical constraints
        reject, revise, or authorise the end with recorded reasons

        estimate physical, actuarial, systemic, and model risk
        allocate risk to insurance, reserves, counterparties, or retention
        verify solvency, coverage limits, exclusions, and appeal rights

        execute or formally close each authorised decision
        settle transactions and update X, R, A, Q, U, and G
        issue a completion certificate if all scoped predicates hold
        reopen the programme if a defined trigger is observed

    RETURN state, certificate, unresolved exceptions, audit log
END PROCEDURE
```

15.3 Complexity and termination

If the programme has m material asymmetries, d decisions, and r evidence objects, basic indexing and reconciliation can be implemented in approximately linear or $O((m+d+r)\log(m+d+r))$ time, depending on data structures and sorting. This is not a complexity bound for the full social process. Investigation,

legal review, adversarial verification, and negotiation can require unbounded human time. The AIE non-decidability result also cautions against promising a general algorithm that decides whether every arbitrary ethics programme will terminate [3]. Software can support bounded workflows; it cannot prove global ethical closure.

15.4 AI and machine learning

Machine learning can assist with anomaly detection, document classification, transaction matching, fraud indicators, claim triage, scenario generation, and estimation of latent state variables. A model may estimate

$$\widehat{L}_i = f_\theta(x_i), \tag{85}$$

where x_i is a feature vector and θ is learned from data. Prediction uncertainty, distribution shift, label error, missingness, and model uncertainty must be incorporated into the risk register. An anomaly score is a reason to investigate, not proof of wrongdoing.

An AI system must not be the sole authority for legal entitlement, insurance denial, credit refusal, employment action, clinical diagnosis, or coercive state action. High-impact outputs require human review, contestability, explanation, independent validation, and monitoring for differential error. Model governance should include data lineage, version control, performance bounds, red-team testing, incident response, and documented change management. NIST's AI Risk Management Framework and Cybersecurity Framework offer useful reference structures [30, 31].

15.5 Cybersecurity and integrity controls

Actually is only as credible as its security model. A threat model should consider:

- falsification or deletion of source records;
- replay, timestamp manipulation, or version substitution;
- unauthorised disclosure of private or commercially sensitive data;
- collusion among an originating party, verifier, and decision authority;
- compromise of signing keys or privileged accounts;
- model poisoning, data drift, and manipulation of training labels;
- denial of service against reporting, claims, or appeals;
- coercion of participants into accepting a false or incomplete record.

Controls include least privilege, multifactor authentication, separation of duties, encryption in transit and at rest, signed records, immutable or append-only audit logs, key rotation, backup and recovery, independent penetration testing, secure software development, and incident reporting. A hash chain can make later alteration detectable under stated cryptographic assumptions; it cannot establish the truth of the original input. Digital signatures establish that a key signed a message; they do not establish that the signer had authority or that the message was accurate. These distinctions are essential to avoid treating technological integrity as epistemic truth [36, 27, 28, 31].

Privacy-preserving computation can support controlled analysis when raw personal data cannot be broadly shared. Differential privacy provides a formal way to bound the effect of an individual record on a released statistic, but privacy noise can reduce the precision of transaction-level verification. A system must disclose that trade-off and should not use a noisy public statistic as if it were an exact audit record [17].

16 Interdisciplinary synthesis

Table 4: Interdisciplinary correspondences and their limits.

Field	**Useful correspondence**	**Limit or safeguard**
Mathematics	Linear invariants, stochastic processes, optimisation, fixed points, and risk allocation.	Formal results depend on explicit assumptions; an analogy is not a proof.
Logic	Separate truth, evidence, inference, decision, and legitimacy predicates.	Syntactic consistency does not establish empirical truth.
Economics	Adverse selection, moral hazard, principal-agent problems, signalling, and market failure.	Disclosure can change incentives and strategic behaviour; institutions must be designed endogenously.
Finance	Balance sheets, value trajectories, capital allocation, liquidity, solvency, and risk pricing.	Book value, market value, economic value, and social value are not interchangeable.
Statistics	Measurement error, posterior uncertainty, covariance, state estimation, stress testing.	Identifiability and data quality bound inference.
Physics	Conservation and symmetry as a model for invariants.	Accounting conservation is not automatically a physical conservation law or Noether charge.
Chemistry	Stoichiometric matrices and conserved moieties.	Economic claims are institutional valuations, not conserved chemical quantities.
Biology	Homeostatic feedback, adaptation, and resilience.	Economies contain rights, power, and strategic agents; they are not organisms.
Linguistics	Common ground, pragmatic context, translation, and shared definitions.	Literal translation does not guarantee identical legal or economic meaning.
Psychology	Bounded attention, loss aversion, trust, framing, and motivated reasoning.	A behavioural indicator is not a diagnosis or proof of misconduct.
Psychiatry and medicine	Consent, confidentiality, clinical evidence, and continuity of care.	Sensitive health data require purpose limitation and professional oversight.
Philosophy	Epistemic justification, public reason, contractualist endorsement, and rights.	Legitimacy cannot be reduced to an optimisation score.
Religion	Stewardship, charity, debt relief, fair dealing, and limits on exploitation.	Traditions differ; comparative examples must not erase doctrinal diversity.
Law	Disclosure duties, contracts, evidence, remedies, due process, and jurisdiction.	The framework does not displace applicable law or legal counsel.
Political science	State capacity, accountability, public finance, and metric manipulation.	Transparency without institutional independence may become propaganda.

Field	Useful correspondence	Limit or safeguard
Geopolitics and diplomacy	Verification regimes, treaty reporting, cross-border risk, and sovereign commitments.	Information-sharing must account for sovereignty, security, and treaty rules.
Welfare economics	Consumption smoothing, distribution, insurance access, and social risk pooling.	Aggregate welfare gains may conceal concentrated losses.
Warfare economics	Correlated losses, logistics, infrastructure resilience, and public-risk finance.	Civilian protection and humanitarian law remain binding.
Computer science	State machines, provenance graphs, workflow engines, and formal verification.	Program correctness does not make inaccurate data true.
AI/ML	Anomaly detection, forecasting, claim triage, and model uncertainty estimation.	Models are decision support, not moral or legal authorities.
Cybersecurity	Integrity, authentication, access control, key management, and incident response.	Integrity of a record is distinct from truth of its content.

17 Institutional design and implementation roadmap

17.1 Minimum viable institutional arrangement

A pilot should begin with one bounded use case rather than an entire national economy. Suitable candidates include a public infrastructure contract, a bank loan with complex collateral, an insurance claims portfolio, a supply-chain finance arrangement, or a development-finance programme. The pilot should identify the participants, material claims, value-state model, legal framework, risk pool, and measurable outcomes before implementation.

The minimum governance arrangement requires: a programme owner; a data steward; an accounting and valuation lead; an independent verifier; an investigation officer; a decision authority; a risk and insurance lead; a privacy and cybersecurity officer; and an appeal or review channel. One person may hold multiple roles in a small pilot, but conflicts must be declared and compensating controls documented.

17.2 Implementation phases

1. **Scoping:** define the programme boundary, parties, materiality rules, legal obligations, and decision rights.
2. **Baseline:** map existing ledgers, disclosures, evidence sources, risk schedules, and dispute history.
3. **Formalisation:** define transaction mappings, value buckets, asymmetry types, convergence rules, and risk measures.
4. **Prototype:** implement an evidence register, reconciliation engine, investigation workflow, risk allocation module, and audit log.
5. **Validation:** test conservation, provenance, access controls, model robustness, decision fairness, and recovery procedures.

6. **Pilot:** operate on real but bounded decisions with independent oversight and a documented fallback process.
7. **Evaluation:** compare outcomes against the baseline and a credible counterfactual where possible.
8. **Scale:** expand only after legal, technical, economic, and distributive performance is demonstrated.

17.3 Performance scorecard

A scorecard should include both process and outcome measures. Process metrics include reconciliation completion, evidence provenance coverage, verification latency, unresolved asymmetry age, decision closure time, audit exceptions, model version compliance, and appeal outcomes. Outcome metrics include financing cost, forecast error, claims dispute rate, recovery time, insolvency risk, service continuity, distribution of uncovered losses, and participant comprehension. No single weighted index should conceal deterioration in a critical dimension.

18 Limitations, failure modes, and safeguards

18.1 Limits of informational symmetry

Perfectly identical information states are neither feasible nor always desirable. Specialised roles require differentiated access; privacy and security restrict disclosure; and some uncertainty is irreducible. The correct target is a sufficiently common, verifiable, and decision-relevant information basis for the specific programme. A residual asymmetry that cannot lawfully or safely be removed should be declared, bounded, and addressed by independent review or contractual protection.

18.2 Strategic behaviour and capture

Participants may manipulate the materiality threshold, choose favourable models, delay disclosure, exploit ambiguous definitions, or capture the verifier. The system therefore needs external audit, rotating reviewers where appropriate, conflict-of-interest rules, whistleblower channels, and penalties for deliberate misrepresentation. A central platform can itself become a source of monopoly power or informational asymmetry; portability and independent inspection are important safeguards.

18.3 Model risk and tail events

The insurance layer cannot guarantee protection against every event. A model can underestimate dependence, ignore regime change, or fail under unprecedented shocks. Risk allocations should include sensitivity analysis, scenario stress tests, reverse stress tests, capital buffers, reinsurance counterparty analysis, and an explicit unmodelled-risk category. The absence of an estimated risk is not evidence that the risk is zero.

18.4 Normative conflict

Affected parties may disagree even after receiving the same evidence. They may have incompatible values, different time horizons, or unequal capabilities. Informational symmetry is therefore necessary for informed deliberation but not sufficient for agreement. AIE's legitimacy conditions and conflict-resolution procedures must operate alongside rights, representation, procedural justice, and lawful adjudication. The system should preserve dissent instead of converting unresolved disagreement into a false consensus.

18.5 Non-decidability and governance humility

AIE's non-decidability result warns that no general algorithm can determine whether every arbitrary ethics programme terminates from its initial description alone [3]. Consequently, an Actually platform must not promise to certify the eventual termination of all ethical programmes. It can only assess

a bounded programme against specified evidence, progress, authority, and closure rules. Governance humility is a technical requirement, not merely a rhetorical virtue.

19 Principal results and research propositions

The integrated theory can be summarised in the following results.

Theorem 19.1 (Conservation of actualised value)**.** *If the initial value state is balanced and every recorded transaction operator preserves the balance functional, then the value-state representation remains balanced throughout the programme.*

Proof. This is the transaction-conservation theorem proved in Section 4. The induction argument is independent of the particular transaction type, provided every transaction is represented with its required counterpart entry. □

Theorem 19.2 (Local epistemic and decisional closure)**.** *If material asymmetry is bounded, investigation is honest and fair, each remaining asymmetry is eventually addressed, and every enabled decision is eventually closed, then a finite programme can reach a state with no unresolved material asymmetry and no outstanding decision.*

Proof. The finite-program convergence theorem in Section 5 applies. The finiteness and progress assumptions provide a decreasing integer potential for the unresolved asymmetry set; decisional closure removes the second termination obstruction. □

Theorem 19.3 (Residual-risk allocation consistency)**.** *If the aggregate risk measure is differentiable and positively homogeneous of degree one, Euler allocations sum exactly to aggregate risk capital. A premium schedule formed from expected indemnity, allocated risk capital, and explicitly stated expenses therefore reconciles at the portfolio level.*

Proof. Euler's homogeneous-function identity gives the risk-capital reconciliation. Summation of the premium components gives the portfolio equation in Section 6. □

Proposition 19.4 (Information and optimal decision quality)**.** *Under a fixed action set and loss function, and when more informative data can be ignored, an agent's optimal ex ante Bayes risk cannot increase when the agent receives a signal that generates a richer information set.*

Proof. The original decision rules remain feasible after the more informative signal is observed. The expanded information set can therefore reproduce the earlier optimal policy or select a better one. The infimum of expected loss cannot increase. □

Corollary 19.5 (The scope of Actually)**.** *Under the assumptions above, Actually can guarantee the stated accounting, programme-closure, legitimacy, and risk-allocation properties for a bounded programme. It cannot guarantee omniscience, eliminate every future asymmetry, or prove the correctness of evidence beyond its validation assumptions.*

20 Conclusion

The three source treatises can be combined because they address different dimensions of the same institutional problem. Value-Based Accounting supplies a dynamic representation of value and a conservation discipline. Asymmetric Informational Ethics supplies the process by which agents investigate what they do not know, enable decisions, and test the legitimacy of their ends. The General Theory of Insurance supplies the machinery for measuring and allocating uncertainty that remains after reasonable investigation.

The synthesis yields a precise conception of Actually: not a new accounting standard, not a universal disclosure mandate, and not a promise to eliminate risk, but a jointly governed technology that makes material economic claims verifiable, disagreements investigable, decisions legitimate, and residual risk explicit. Its central institutional rule is simple: no material value claim should be acted upon without an

adequate record; no material asymmetry should be ignored without a reasoned disposition; no decision should be treated as legitimate merely because it is informed; and no residual risk should be transferred without identifying who bears it.

The Egyptian–Roman problem is thus transformed into a cooperative design problem. Accounting can create a common representation, but only joint verification, investigation, and risk-sharing can make that representation mutually usable. The spanner economy adds an important implication: technological capability without accounting does not by itself provide a complete basis for mutual economic understanding. Actually connects productive capability to verifiable value and accountable cooperation.

The proposed theory is ambitious, but its claims are deliberately bounded. Mathematical results depend on stated assumptions; ethical legitimacy requires public justification; legal validity depends on jurisdiction and applicable law; actuarial reliability depends on data and model quality; and empirical success must be demonstrated rather than presumed. The next stage is to formalise a reference implementation, test it on bounded institutional settings, and compare its performance against existing accounting, audit, disclosure, and insurance arrangements.

Actually is the technology through which value becomes verifiable, information becomes mutually actionable, decisions become justifiable, and residual uncertainty becomes accountable.

A Notation and formal checklist

Table 5: Principal notation.

Symbol	Meaning
$\mathcal{N}$	Set of agents or participants.
$I_i(t)$	Information state of agent i at time t.
$\mathcal{A}_\Omega(t)$	Total pairwise informational asymmetry field.
$\mathcal{A}_P^{\mathrm{mat}}(t)$	Material, programme-relevant asymmetry set.
$\mathcal{A}_P^{\mathrm{open}}(t)$	Open material asymmetries not yet resolved or validly dispositioned.
P	Bounded ethical or institutional programme.
E_P	Intended end of a programme.
Φ	Investigation function.
D_0	Set of decisions that may be enabled.
$Q_P(t)$	Decisions still outstanding at time t.
$A(t), L(t), E(t)$	Aggregate assets, liabilities, and equity.
$X(t)$	Nine-component value-state vector.
b	Balance functional vector.
$M(t)$	Value momentum, equal to value creation less value destruction.
L_i	Loss associated with exposure i.
ρ	Aggregate risk measure.
c_i	Euler allocation of aggregate risk capital to exposure i.
P_i	Insurance premium for exposure i.
Z_t	Integrated Actually system state.
certificate	Portable record of evidence, decisions, risk allocation, and governance.

B Actually programme checklist

Before a programme is certified, an independent reviewer should be able to answer the following questions:

1. Is the programme boundary explicit, including relevant parties, time horizon, and materiality threshold?
2. Are value states and transaction flows reconciled, including boundary and contingent accounts?
3. Can every material claim be traced to a source, responsible actor, version, and verification status?
4. Have material informational differences been identified, investigated, and either resolved or explicitly excepted?
5. Are all decisions recorded as made, rejected, deferred under a rule, or still outstanding?
6. Does the proposed end satisfy the applicable legal constraints and the adopted legitimacy test?
7. Are model, actuarial, systemic, physical, operational, legal, cyber, and geopolitical risks separately considered?
8. Does each residual risk have an identified bearer, reserve, insurance arrangement, or explicit retention decision?
9. Are privacy, cybersecurity, due process, appeal, and independent review arrangements adequate?
10. Are reopening triggers specified for new evidence, material model error, fraud, or changed circumstances?

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Glossary

Actually
: The integrated institutional technology for verifiable value, investigation of material informational asymmetries, legitimate decision, and residual-risk allocation.

Accounting conservation
: The requirement that recognised value changes be represented through balanced transaction entries and counterpart accounts.

Asymmetric information
: A condition in which agents hold materially different information about a decision-relevant proposition.

Bayes risk
: The minimum expected decision loss available under a specified information set, action set, and loss function.

Decision closure
: The state in which every enabled decision has been executed, rejected, superseded, or otherwise disposed of under an authorised rule.

Evidence provenance
: The recorded origin, time, author, version, custody, and verification history of an evidence object.

Ideal endorsement
: The AIE condition that an end would be endorsable by affected parties under the specified full-information condition.

Material asymmetry
: An informational difference with sufficient decision relevance and materiality to warrant investigation within a programme; its open status ends only through verified resolution or an authorised substitute disposition.

Model risk
: Uncertainty arising from competing or misspecified models, parameters, assumptions, or estimation procedures.

Residual risk
: Risk that remains after reasonable investigation and information improvement.

Risk allocation
: The assignment of measured risk or capital contributions to participants, insurers, counterparties, or retained-risk accounts.

Value momentum
: The difference between expected value-creating and value-destroying flows in the Value-Based Accounting framework.

Value-state vector
: The nine-component representation of increasing, constant, and decreasing assets, liabilities, and equity.

The End